

Species-Aware Review: RecBCD End Resection in *Pseudomonas putida* KT2440

Module/bucket: `recbcd_end_resection` (mapped to KEGG `ppu03440`, Homologous recombination) **Target taxon:** *Pseudomonas putida* KT2440 — PSEPK, NCBI taxon 160488, proteome UP000000556 **Review date:** 2026-08-08 · Iteration 1

1. Executive summary

The RecBCD end-resection step of homologous recombination (HR) is **satisfiable and COVERED** in *P. putida* KT2440. All three canonical subunits are encoded in a **contiguous, co-oriented recD–recB–recC operon** — **PP_4672 (Q88DZ6) · PP_4673 (Q88DZ5) · PP_4674 (Q88DZ4)** — with subunit lengths matching functional *E. coli* RecBCD and the expected catalytic domains (RecB: UvrD-helicase + PDDEXK nuclease; RecC: Exonuc_V_gamma; RecD: RecD helicase). The candidate list also correctly captures the downstream/parallel HR core (RecA, RecFOR, RecJ, RuvABC, RecG, PriA).

Three curation-critical caveats:

1. **Every RecBCD annotation is homology-inferred (UniProt PE3), with no direct experimental evidence in KT2440.** Presence is confident; mechanistic detail is transferred.
2. **The end-resection *regulatory* mechanism does not transfer from *E. coli*.** Direct experiments show *P. putida* RecBCD-like enzyme has nuclease/helicase activity **but does not recognize the *E. coli* Chi site (5'-GCTGGTGG-3')** and shows **no Chi hotspot activity**

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Any Chi-based annotation is an over-propagation.

3. **Gene presence ≠ pathway proficiency.** KT2440 is intrinsically DNA-damage-sensitive with **poor HR efficiency and a weak SOS response**, traced to an inefficient RecA–LexA interplay

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Several resection-relevant genes are **absent from the 24-gene metadata** but present in the proteome and should be added to the step model: **recN**, **sbcCD**, **sbcB (ExoI)**, **xseA (ExoVII)**, **recQ**, **radA**, plus an unexpected **Ku/LigD NHEJ** system.

2. Target-organism pathway definition

Process included (`recbcd_end_resection`): the *initiation* of RecA-dependent HR at a double-strand DNA break (DSB). RecBCD binds a blunt/near-blunt duplex end and couples ATP-driven unwinding (RecB 3'→5' and RecD 5'→3' motors) to nucleolytic degradation, ultimately generating a 3'-tailed ssDNA onto which RecA is loaded. In *E. coli* this switch is triggered by the Chi octamer; the equivalent trigger in *Pseudomonas* is unknown (see §5).

Boundaries — keep separate from: - **RecFOR/RecJ gap repair & backup resection** (recF, recO, recR, recJ, recQ): initiates HR at ssDNA gaps and substitutes for RecBCD nuclease/loading when RecBCD is inactivated. Mechanistically adjacent, but a *distinct* initiation route — not part of RecBCD end resection proper. - **Branch migration / resolution** (ruvABC, recG): post-synaptic, downstream of resection. - **Replisome / replication-restart** (dnaN, dnaE, dnaQ, dnaX, holABC, ssb, polA, priA): KEGG ppu03030/03430 overlap. These co-occur in the candidate list because ppu03440 shares components, but they are not resection enzymes (priA and ssb are shared cofactors). - **NHEJ** (ku, ligD): an independent, RecA-independent DSB pathway.

Alternate names / DB definitions: Exonuclease V (ExoV); "Helicase/nuclease RecBCD"; EC 3.1.11.5 (RecB nuclease), EC 5.6.2.3 (RecD helicase), EC 5.6.2.4 (RecB helicase). In Firmicutes the functional analog is **AddAB (RexAB)** — *absent* here, as expected for a gammaproteobacterium.

3. Expected step model (RecBCD end resection)

Step	Enzyme/function	KT2440 gene	Status
Duplex-end binding & unwinding	RecBCD helicase (dual motor)	recB, recC, recD	Covered
3'→5' / 5'→3' nucleolysis	RecB nuclease domain	recB	Covered
Recombinogenic switch at hotspot	(Chi in <i>E. coli</i> ; unknown in <i>Pseudomonas</i>)	—	Gap / needs revision
RecA loading onto 3' ssDNA	RecBCD (or RecFOR backup)	recB/recC + recFOR	Covered (mechanism uncertain)
Backup/alternative resection	RecJ 5'-exo + RecQ helicase + SSB	recJ, recQ, ssb	Covered (parallel path)
End trimming / hairpin cleavage	SbcCD, ExoI, ExoVII	sbcC/sbcD, sbcB, xseA	Present, not in metadata

4. Candidate genes and evidence

High-confidence RecBCD core (promote-worthy): - **recB** — PP_4673 / Q88DZ5 (1224 aa). 3'-5' helicase + single nuclease active site; the catalytic heart of resection and RecA loading. EC 3.1.11.5 / 5.6.2.4. Homology annotation; length consistent with *E. coli* RecB (~1180 aa). **Curation caveat:** the nuclease/RecA-loading regulation is Chi-independent in *Pseudomonas* — do not transfer *E. coli* Chi mechanism. - **recC** — PP_4674 / Q88DZ4 (1160 aa). Scaffolding/Chi-recognition subunit in *E. coli*; in *Pseudomonas* the Chi-reader function is presumed non-functional or redirected to an unknown sequence. Flag GO terms implying Chi recognition. - **recD** — PP_4672 / Q88DZ6 (691 aa). 5'-3' motor; loss converts RecBCD to a Chi-independent recombinase in *E. coli*. Homology only.

Correctly assigned adjacent HR genes: recF (Q88RW7), recO (Q88MY3), recR (Q88F32), recJ (Q88MU1) — RecFOR/RecJ backup route; recA (Q88ME4) — central recombinase; ruvA/ruvB/ruvC (Q88NJ1/0/2), recG (Q88C73) — Holliday-junction migration/resolution (downstream, not resection); priA (Q88CU2) — replication restart (shared cofactor).

Likely over-inclusions for a *resection* module (replisome/other buckets): dnaN, polA, ssb, holC, dnaEA, holB, dnaQ, dnaX, holA, and the RNase-T exonucleases **PP_0353 / PP_4768** (see below). These belong to ppu03030/03420 replication; **ssb** is a genuine shared HR cofactor, the DNA-Pol-III subunits and RNase-T exonucleases are not resection enzymes and should not count toward RecBCD-step satisfiability.

Annotation quality flag: all 13 HR genes checked are **UniProt PE3 "Inferred from homology"** — none has target-strain experimental protein evidence.

Resolved (Iteration 2): **PP_0353** (Q88QY1, 236 aa) and **PP_4768** (Q88DQ5, 203 aa) both carry **Pfam PF00929 (RNase_T) / RNase-H-like fold (SSF53098)** — the **DEDDh 3'-5' exonuclease family** (RNase T / oligoribonuclease / DnaQ-epsilon-like proofreading). PP_4768's own UniProt function text describes Pol-III-epsilon-type proofreading activity. **Neither is a RecBCD-pathway resection nuclease** — they should be **EXCLUDED** from this module, not promoted.

5. Gaps, ambiguities, and likely over-annotations

- **Chi-site over-propagation (highest priority).** Direct evidence

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P. putida/*P. aeruginosa* RecBCD confers recombination and ATP-dependent nuclease activity **but no Chi hotspot activity and no Chi-dependent cleavage.** → Remove/annotate-as-not-applicable any 5'-GCTGGTGG recognition claim; the *Pseudomonas* recombination-hotspot sequence is an **open question**.

- **Missing resection genes in metadata:** recN (Q88DU0), sbcC (Q88LB1)/sbcD (Q88LB0), sbcB/ExoI (Q88N51), xseA/ExoVII (Q88P26), recQ (Q88EE9), radA (Q88E24). These are bona fide end-processing / recombination factors and should be represented in the step model.
- **Unexpected pathway present: Ku (Q88HU8) + LigD (Q88HU3) NHEJ** — a RecA-independent DSB route absent from *E. coli*. Relevant context when reasoning about DSB-repair satisfiability, though outside the resection step itself.
- **Phenotype vs. genotype mismatch:** complete gene set but **poor HR / weak SOS**

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Functional inference from presence alone is unsafe.

- **AddAB not expected** and correctly absent — do not open an AddAB gap for this taxon.

6. Module and GO-curation recommendations

Module step	Recommendation
RecBCD binding / unwinding / nucleolysis / RecA loading	covered (recB/recC/recD present, contiguous, correct length)
Chi-triggered recombinogenic switch	module_needs_revision / candidate_uncertain — <i>E. coli</i> Chi model does not apply; flag as gap for the hotspot signal
Backup resection (RecJ/RecQ/RecFOR)	covered , but keep as a <i>separate</i> initiation arm from RecBCD
SbcCD / ExoI / ExoVII end trimming	covered but add to metadata (currently missing)
AddAB alternative	not_expected_in_target_taxon

Boundary fix: the generic bucket conflates ppu03440 (HR) with ppu03030 (replication) genes. For a `recbcd_end_resection` module, restrict counted genes to recB/recC/recD (+ optionally recA, ssb, recJ, recQ as cofactors) and exclude DNA-Pol-III subunits.

GO/annotation requests: (i) a note or evidence flag that KT2440 RecBCD is **Chi-independent**; (ii) add recN, sbcC/sbcD, recQ, sbcB, xseA to the resection step; (iii) resolve broad "Exonuclease" mappings for PP_0353 and PP_4768 before module inclusion.

7. Genes to promote to full `fetch-gene` review

1. **recB (PP_4673)** — catalytic core; verify nuclease/helicase domains and the Chi-independence caveat.
2. **recC (PP_4674)** — confirm whether the Chi-recognition tunnel is degenerate/redirected in *Pseudomonas*.
3. **recD (PP_4672)** — confirm 5'-3' motor and operon context.
4. **recJ (PP_1477) + recQ (Q88EE9)** — define the backup-resection arm and its metadata placement.

5. ~~PP_0353 and PP_4768~~ — **resolved in Iteration 2: RNase T/DEDDh-family (PF00929) proofreading exonucleases, NOT RecBCD resectors. Exclude from the module; no fetch-gene needed for this module.**

Domain confirmation (Iteration 2): RecB carries UvrD-helicase (PF00580) + PDDEXK nuclease (PF12705) domains; RecC carries the Exonuc_V_gamma domain (PF04257); RecD carries the RecD/UvrD-like helicase core — all three subunits have the expected catalytic architecture, reinforcing the COVERED call.

8. Key references

- McKittrick NH, Smith GR. *Activation of Chi recombinational hotspots by RecBCD-like enzymes from enteric bacteria.* J Mol Biol 1989.

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— Direct evidence that *P. putida*/*P. aeruginosa* RecBCD lacks *E. coli* Chi recognition.

- Akkaya Ö, Aparicio T, Pérez-Pantoja D, de Lorenzo V. *The faulty SOS response of Pseudomonas putida KT2440 stems from an inefficient RecA-LexA interplay.* 2021.

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— Target-strain HR/SOS deficiency.

- Vlašić I et al. *The hybrid recombinational repair pathway operates in a χ -deficient recC1004 mutant of E. coli.* 2012.

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— RecJ/RecFOR substitute for RecBCD nuclease/loading (backup-resection logic).

- Lenhart JS et al. *RecO and RecR are necessary for RecA loading...* 2014.

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— RecFOR mediator role (transfer: strong, conserved).

- Gupta R, Shuman S, Glickman MS. *RecF and RecR play critical roles in HR and SSA of mycobacteria.* 2015.

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— Comparative caveat on RecF-pathway variability across taxa.

- UniProt proteome **UP000000556** (accessions in §4) — annotation source; all HR entries are PE3 "Inferred from homology".

Evidence basis: Gene presence/lengths/operon context are direct from the KT2440 reference proteome (authoritative genomic evidence). The Chi-independence and HR/SOS-weakness conclusions are **direct experimental evidence in *P. putida*** (PMIDs 2559208, 33393180). RecFOR/RecJ mediator mechanisms are transferred from *E. coli*/*B. subtilis*/mycobacteria (strong conservation, but no KT2440-specific assay). No functional RecBCD biochemistry has been published for KT2440 specifically — the largest open gap.

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